

## Signalling game

Consider the following signalling game. Nature chooses the worker's type: high ability  $t_A$  with probability  $\frac{1}{4}$  and low ability  $t_B$  with probability  $\frac{3}{4}$ . The worker observes his type and chooses an education level: entering the labor market with an undergraduate degree  $U$ , or obtaining a graduate degree  $G$ . The education costs are

$$c(U, t_A) = c(U, t_B) = 0 \quad c(G, t_A) = 2 \quad c(G, t_B) = 5$$

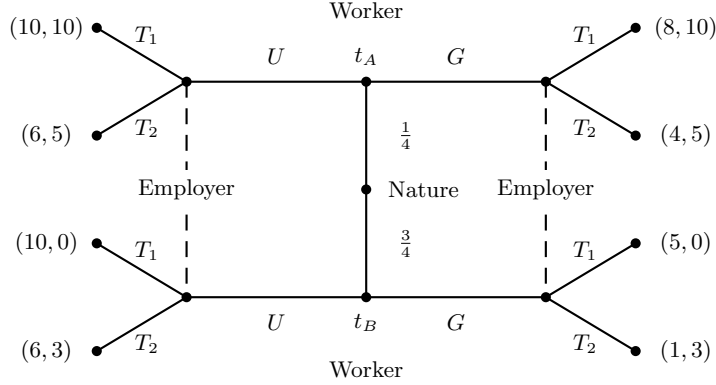
The employer observes education, but not type, and offers either job  $T_1$  or job  $T_2$ . Wages are 10 under  $T_1$  and 6 under  $T_2$ . The worker's payoff is wage minus education cost. The employer's payoff is

$$10 \text{ if } (t_A, T_1) \quad 5 \text{ if } (t_A, T_2) \quad 0 \text{ if } (t_B, T_1) \quad 3 \text{ if } (t_B, T_2)$$

1. Draw the game in extensive form and indicate what kind of game it is
2. Determine which separating equilibria are Perfect Bayesian equilibria
3. Determine which pooling equilibria are Perfect Bayesian equilibria
4. Interpret your results from parts b) and c). Do all of them make economic sense

## Solution

1. This is a dynamic game of incomplete information. More specifically, it is a signalling game: the worker knows his own type and sends an education signal to the employer, who then updates beliefs and chooses a job offer



**Conclusion:** this is a signalling game, that is, a dynamic game of incomplete information

2. Let  $\mu$  be the employer's posterior belief that the worker is high ability after observing an education signal. If the employer chooses  $T_1$ , expected payoff is

$$\Pi_E(T_1 | \mu) = 10\mu$$

If the employer chooses  $T_2$ , expected payoff is

$$\Pi_E(T_2 | \mu) = 5\mu + 3(1 - \mu) = 3 + 2\mu$$

Hence,  $T_1$  is optimal if and only if

$$10\mu \geq 3 + 2\mu \iff 8\mu \geq 3 \iff \mu \geq \frac{3}{8} = 0,375$$

Now consider the separating candidates.

**Separating candidate 1:**  $t_A \rightarrow G$  and  $t_B \rightarrow U$

Bayes' rule gives

$$\mu(G) = 1 \quad \mu(U) = 0$$

Therefore the employer chooses  $T_1$  after  $G$  and  $T_2$  after  $U$

Worker incentives:

$$\begin{aligned} u_A(G, T_1) &= 10 - 2 = 8 & u_A(U, T_2) &= 6 - 0 = 6 \\ u_B(U, T_2) &= 6 - 0 = 6 & u_B(G, T_1) &= 10 - 5 = 5 \end{aligned}$$

So neither type wants to deviate.

**Separating candidate 2:**  $t_A \rightarrow U$  and  $t_B \rightarrow G$

Bayes' rule gives

$$\mu(U) = 1 \quad \mu(G) = 0$$

Therefore the employer chooses  $T_1$  after  $U$  and  $T_2$  after  $G$

Now the low type obtains

$$u_B(G, T_2) = 6 - 5 = 1 \quad u_B(U, T_1) = 10 - 0 = 10$$

Hence the low type deviates.

**Conclusion:** the only separating Perfect Bayesian equilibrium is

$$t_A \rightarrow G \quad t_B \rightarrow U \quad G \rightarrow T_1 \quad U \rightarrow T_2$$

with beliefs

$$\mu(G) = 1 \quad \mu(U) = 0$$

3. Now consider pooling equilibria.

**Pooling candidate 1:** both types choose  $U$

Then Bayes' rule implies

$$\mu(U) = \frac{1}{4} = 0,25$$

Since

$$\frac{1}{4} < \frac{3}{8}$$

the employer chooses  $T_2$  after  $U$

The signal  $G$  is off the equilibrium path, so  $\mu(G)$  is free. To deter deviations we need the employer to choose  $T_2$  after  $G$ , which requires

$$\mu(G) \leq \frac{3}{8} = 0,375$$

Deviation checks:

$$\begin{aligned} u_A(U, T_2) &= 6 - 0 = 6 & u_A(G, T_2) &= 6 - 2 = 4 \\ u_B(U, T_2) &= 6 - 0 = 6 & u_B(G, T_2) &= 6 - 5 = 1 \end{aligned}$$

So neither type wants to deviate.

**Pooling candidate 2:** both types choose  $G$

Then Bayes' rule implies

$$\mu(G) = \frac{1}{4} = 0,25$$

Hence the employer chooses  $T_2$  after  $G$ , and the high type obtains

$$u_A(G, T_2) = 6 - 2 = 4$$

If the high type deviates to  $U$ , even in the least favorable case in which the employer also chooses  $T_2$ , he obtains

$$u_A(U, T_2) = 6 - 0 = 6$$

Thus the high type deviates.

**Conclusion:** the pooling Perfect Bayesian equilibria are exactly those in which both types choose  $U$ , the employer chooses  $T_2$  after  $U$  and also  $T_2$  after  $G$ , with beliefs

$$\mu(U) = \frac{1}{4} = 0,25 \quad \mu(G) \leq \frac{3}{8} = 0,375$$

There is no pooling Perfect Bayesian equilibrium in which both types choose  $G$

4. The separating equilibrium has a clear economic interpretation: graduate education works as a costly signal. The high-ability worker can obtain the graduate degree at relatively low cost, while the low-ability worker finds imitation too expensive.

The pooling equilibrium on  $U$  means that extra education conveys no information in equilibrium, so the employer offers the lower-level job  $T_2$  to everyone. This outcome is possible only because the off-path belief after  $G$  is sufficiently pessimistic, namely  $\mu(G) \leq \frac{3}{8}$

Thus, not all equilibrium outcomes are equally convincing economically. The separating equilibrium is the most natural one. The pooling equilibrium on  $U$  is technically valid as a Perfect Bayesian

equilibrium, but it relies on off-path beliefs that make the employer interpret graduate education as not informative enough. The rejected candidates do not make economic sense as equilibria because at least one type has a profitable deviation.

**Final conclusion:** the economically most plausible outcome is the separating equilibrium in which the high type chooses  $G$  and the low type chooses  $U$ . A pooling equilibrium on  $U$  also exists, but it is sustained only by restrictive off-path beliefs